# The lognormal distribution.

**Definition 1.1.** Let $X \sim Normal(mean = m, variance = \nu^2)$. Define the random variable $Y = e^X$. We say that the random variable $Y$ is *lognormally distributed*.

## 1.1. First properties.

- The expected value of the lognormally distributed random variable $Y$ can be obtained as follows:
$$\mathbb{E}[Y] = \mathbb{E}[e^X] = M_X(1) = e^{m+\frac{\nu^2}{2}}.$$
- Let $Y$ be a lognormal and let $a \neq 0$. Then, the random variable $Y^a$ is also lognormal. *Note:* For $a = 0$, we get a degenerate random variable at 1 which can, technically, be interpreted as lognormal, but is not fun.
- Let $Y_1$ and $Y_2$ be independent and lognormally distributed. Then, $Y_1 Y_2$ is also lognormal.

## 1.2. Quantiles.

**Definition 1.2.** For $p$ such that $0 < p < 1$, we define the $100p^{th}$ *quantile* of a random variable $X$ as any value $\pi_p$ such that
$$F_X(\pi_p-) \leq p \leq F_X(\pi_p).$$
In particular, the $50^{th}$ quantile is referred to as the *median*.

*Note:* When the random variable $X$ is continuous, we can obtain the $100p^{th}$ quantile by simply solving for $\pi_p$ in
$$F_X(\pi_p) = p.$$
Consider a probability $p$. Let $z_p$ be the $100p^{th}$ quantile of the standard normal distribution. Let $Y$ be lognormally distributed as above. My claim is that the value
$$y_p = e^{m+\nu z_p}$$
is the $100p^{th}$ quantile of $Y$. Let us simply verify this claim by calculating $F_Y(y_p)$. We have, with $Z \sim N(0,1)$,
$$F_Y(y_p) = \mathbb{P}[Y \leq y_p] = \mathbb{P}[e^X \leq y_p] = \mathbb{P}[e^{m+\nu Z} \leq e^{m+\nu z_p}].$$
Since the logarithmic function is increasing, we have that the above equals
$$F_Y(y_p) = \mathbb{P}[m + \nu Z \leq m + \nu z_p] = \mathbb{P}[Z \leq z_p] = p.$$
The above concludes our proof.

In particular, since the median of the standard normal distribution equals 0, the median of the lognormal distribution will be $e^m$.

*Note:* Since
$$e^m < e^{m+\frac{\nu^2}{2}}, \tag{1.1}$$
i.e., since the mean of a lognormal distribution always exceeds the median, we say that it's *right-skewed*. In fact, this is what its probability density function looks like.

# Jensen's Inequality.

Caveat: $\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}$

$X \sim \text{Normal}(\text{mean} = m, \text{var} = v^2)$

**Theorem.** Let $X$ be a random variable, and let $g$ be a convex f'tion such that

$g(X)$ is well-defined and $\mathbb{E}[g(X)]$ exists.

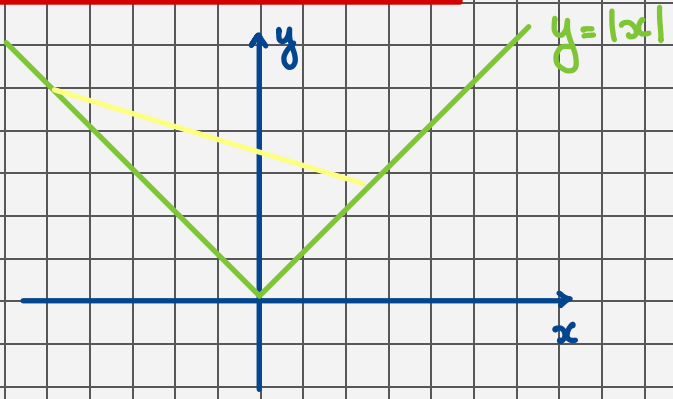
Then,

$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$$

**Example.**

i. $g(x) = |x|$

$$\mathbb{E}[|X|] \geq |\mathbb{E}[X]|$$




ii. Consider a European put w/ strike $K$.

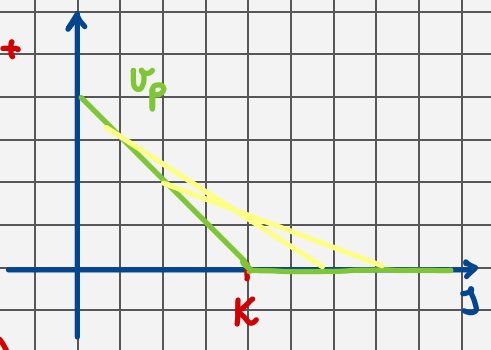
Its payoff f'tion: $v_p(s) = (K-s)_+$

The expected payoff:

$$\mathbb{E}[v_p(S(T))] = \mathbb{E}[(K-S(T))_+]$$

By Jensen's Inequality it's

$$\text{expected payoff} \geq (K - \mathbb{E}[S(T)])_+$$

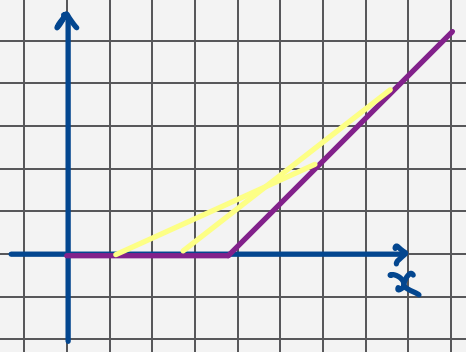


iii. In classical insurance:

$\begin{cases} X \dots \text{(ground-up) loss, i.e., severity r.v.} \\ d \dots \text{deductible} \end{cases}$

The insurer pays

$$(X-d)_+, \text{ i.e., } g(x)=(x-d)_+$$




By Jensen's Inequality:

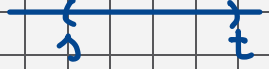
$$\mathbb{E}[(X-d)_+] \geq (\mathbb{E}[X]-d)_+$$

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## Log-Normal Stock Price.

Temporarily, fix a time horizon $T$.

$S(t)$, $t \in [0,T]$ ... time-$t$ stock price




Recall:

$$R(s,t) := \ln\left(\frac{S(t)}{S(s)}\right) \iff S(t) = S(s)\cdot e^{R(s,t)}$$

In particular: $R(0,T)$ ... realized return for $[0,T]$

**We model realized returns as normal.**

$$R(0,T) \sim \text{Normal}(\text{mean}=m, \text{var}=v^2)$$

$\Rightarrow S(T)$ is lognormal

$$\Rightarrow \mathbb{E}^*[S(T)] = \mathbb{E}^*[S(0)e^{R(0,T)}] = \boxed{S(0)\,e^{m+\frac{v^2}{2}}} \quad (\star)$$

Market Model:

- Riskless asset: w/ ccrfir $r$
- Risky asset: a non-dividend-paying stock w/ volatility $\sigma$

Q: Under the risk-neutral $\mathbb{P}^*$, we have

$$\mathbb{E}^*[S(T)] = S(0)e^{rT} \quad (\star\star)$$

Equating $(\star)$ & $(\star\star)$, we get: $m + \frac{v^2}{2} = rT$

Q: $\mathrm{Var}[R(0,1)] = \sigma^2$

$$\Rightarrow \mathrm{Var}[R(0,T)] = \sigma^2 \cdot T = v^2$$

$$m = rT - \frac{1}{2}\sigma^2 T = (r - \tfrac{1}{2}\sigma^2)\cdot T$$

$$\Rightarrow R(0,T) \sim \text{Normal}(\text{mean} = (r - \tfrac{\sigma^2}{2})\cdot T,\ \text{var} = \sigma^2 \cdot T)$$

Say, $Z \sim N(0,1)$

Then, $R(0,T)$ can be expressed as

$$R(0,T) = (r - \tfrac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z$$

Then,

$$S(T) = S(0)e^{R(0,T)} = S(0)e^{(r - \frac{\sigma^2}{2})\cdot T + \sigma\sqrt{T}\cdot Z}$$

Q: What is the median of $S(T)$ under $\mathbb{P}^*$?

$$S(0)\, e^{(r - \frac{\sigma^2}{2})\cdot T}$$